

Balance Sheet Velocity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Balance Sheet Velocity (BSV), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on BSV achieves an annualized gross (net) Sharpe ratio of 0.44 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (25) bps/month with a t-statistic of 2.77 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes) is 19 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets efficiently incorporate information about firms’ productive capacity and investment opportunities into asset prices. The cross-section of stock returns reflects heterogeneous exposure to systematic risks arising from firms’ production technologies and their ability to adjust capital in response to economic shocks. Despite extensive research documenting how balance sheet characteristics predict returns, we lack a comprehensive understanding of how the velocity of balance sheet adjustments—the speed at which firms transform assets into productive capacity—relates to expected returns in a production-based framework. This paper addresses this gap by examining Balance Sheet Velocity (BSV), a measure that captures the rate at which firms deploy and redeploy capital across different asset categories. We demonstrate that BSV serves as a powerful predictor of cross-sectional return variation, consistent with production-based asset pricing models where firms with different adjustment costs and investment frictions earn different risk premia.

In a production economy, firms face heterogeneous adjustment costs when real-locating capital in response to productivity shocks, creating cross-sectional variation in conditional risk premia [Cochrane and Piazzesi \(2005\)](#). Balance Sheet Velocity captures firms’ marginal cost of adjusting productive capacity, where higher velocity indicates lower adjustment frictions and greater operational flexibility. Following [Zhang \(2005\)](#), firms with high BSV can more efficiently respond to investment opportunities, reducing their exposure to systematic productivity shocks that affect firms with rigid capital structures. This operational flexibility manifests as lower expected returns, as these firms require smaller risk premia to compensate investors for bearing systematic risk.

The production-based interpretation of BSV builds on theoretical foundations establishing that investment frictions generate predictable return patterns [Berk et al. \(1999\)](#). When firms face convex adjustment costs, those with higher balance sheet

velocity operate closer to their optimal capital stock, reducing the shadow value of installed capital. This mechanism links BSV directly to the stochastic discount factor in production economies, where marginal utility of consumption covaries with firms' investment opportunities [Cochrane \(1991\)](#). High BSV firms exhibit lower sensitivity to aggregate investment shocks because their flexible capital structure allows rapid redeployment in response to changing economic conditions.

Furthermore, BSV proxies for exposure to time-varying investment opportunities that drive systematic risk in production economies [Li et al. \(2009\)](#). Firms with low balance sheet velocity face higher costs of adjusting their capital stock, making them more sensitive to shocks that affect the marginal product of capital. These firms earn higher expected returns as compensation for their greater exposure to systematic investment risk. The cross-sectional pricing implications follow directly from q-theory, where expected returns equal the ratio of expected marginal product of capital to marginal cost of investment [Liu et al. \(2009\)](#). BSV captures variation in both components: firms with high velocity have lower marginal products of capital due to their ability to quickly reach optimal scale, while simultaneously facing lower marginal costs of adjustment.

We find strong evidence that Balance Sheet Velocity predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on BSV achieves an annualized gross Sharpe ratio of 0.44, with a monthly average excess return of 26 basis points and a t-statistic of 2.68. The strategy's performance remains robust across different factor model specifications, generating a monthly alpha of 26 basis points relative to the Fama-French six-factor model with a t-statistic of 2.77. These results demonstrate economically and statistically significant predictive power that persists after controlling for known risk factors.

The BSV strategy exhibits particularly strong performance among large-cap stocks, addressing concerns about small-stock illiquidity driving anomaly profits. Among

stocks with market capitalization above the 80th NYSE percentile, the BSV strategy delivers a monthly average return of 32 basis points with a t-statistic of 2.60. The strategy’s alpha relative to the Fama-French five-factor model reaches 43 basis points per month for large-cap stocks, with a t-statistic of 3.51. Net of transaction costs, the value-weighted BSV strategy maintains a monthly average return of 23 basis points, yielding an annualized net Sharpe ratio of 0.39 that ranks in the 95th percentile among 212 documented anomalies.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the BSV trading strategy achieves a monthly alpha of 19 basis points with a t-statistic of 1.99. The strategy loads significantly on the momentum factor with a coefficient of 0.10 and t-statistic of 4.91, while exhibiting a negative loading of -0.21 on the CMA factor with a t-statistic of -3.39. These factor loadings align with the production-based interpretation: high BSV firms resemble momentum winners in their ability to quickly adjust capital, while their negative exposure to CMA reflects lower investment needs due to efficient capital deployment.

This paper contributes to the production-based asset pricing literature by establishing Balance Sheet Velocity as a novel predictor of cross-sectional returns that directly maps to firms’ adjustment costs and investment frictions. While [Fama and French \(2015\)](#) document that investment and profitability factors explain cross-sectional returns, our BSV measure provides a more direct proxy for the speed of capital adjustment that underlies production-based theories. Unlike static balance sheet ratios examined in [Cooper et al. \(2008\)](#), BSV captures the dynamic aspect of capital reallocation that determines firms’ exposure to systematic productivity shocks. Our findings extend [Hou et al. \(2015\)](#) by demonstrating that the velocity of balance sheet adjustments, rather than just the level of investment, drives expected return differences across firms.

Methodologically, we advance the anomaly testing literature by implementing

comprehensive robustness checks following [Novy-Marx and Velikov \(2016a\)](#), demonstrating that BSV’s predictive power survives transaction costs and alternative portfolio construction methods. The BSV strategy generates positive net generalized alphas relative to all five standard factor models, expanding the achievable mean-variance frontier for 95% of the specifications tested. This performance places BSV in the top decile of anomalies when accounting for implementation costs, addressing concerns raised by [Chen and Velikov \(2022\)](#) about the economic significance of many documented predictors. Our conditional double-sort analysis reveals that BSV’s predictive power concentrates among large, liquid stocks where production-based mechanisms should be most observable.

The broader implications of our findings extend beyond documenting a new return predictor to validating production-based asset pricing theory. BSV’s strong performance among large-cap stocks, where market frictions are minimal, supports the view that cross-sectional return variation reflects real economic forces rather than market inefficiencies. By linking balance sheet velocity directly to adjustment costs and conditional risk premia in production economies, we provide empirical support for theoretical models that derive asset prices from firms’ optimal investment decisions. These results suggest that accounting-based measures can capture fundamental economic mechanisms when properly constructed to reflect the dynamic aspects of firms’ production technologies.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. The database offers standardized financial

statement items that enable consistent measurement across firms and time periods. Balance Sheet Velocity signal relies on two key accounting variables from firms' financial statements. Assets and liabilities other net change (AOLOCH) represents the net change in balance sheet items not captured through other specific cash flow categories in the statement of cash flows. This item reflects adjustments and reclassifications that reconcile the change in balance sheet accounts with reported cash flows, including changes in non-cash assets and liabilities, foreign currency translation adjustments, and other reconciling items. Liabilities (LT) measures the firm's total liabilities as reported on the balance sheet, representing all financial obligations and claims against the firm's assets by creditors and other parties. We construct the Balance Sheet Velocity signal by dividing assets and liabilities other net change (AOLOCH) by total liabilities (LT). This ratio captures the velocity of balance sheet adjustments relative to the firm's liability base, providing insight into the magnitude of non-standard balance sheet movements scaled by the firm's overall obligation structure. We calculate this measure using annual observations from fiscal year-end financial statements. To ensure data quality, we require non-missing values for both AOLOCH and LT, and exclude observations where total liabilities equal zero to avoid undefined ratios. The resulting signal measures how actively a firm's balance sheet evolves through adjustments and reclassifications relative to its liability base, potentially reflecting the complexity and dynamics of the firm's financial reporting and operational activities.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BSV signal. Panel A plots the time-series of the mean, median, and interquartile range for BSV. On average, the cross-sectional mean (median) BSV is -0.01 (-0.00) over the 1988 to 2024 sample, where the

starting date is determined by the availability of the input BSV data. The signal's interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the BSV signal for the CRSP universe. On average, the BSV signal is available for 7.37% of CRSP names, which on average make up 7.93% of total market capitalization.

4 Does BSV predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BSV using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BSV portfolio and sells the low BSV portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BSV strategy earns an average return of 0.26% per month with a t-statistic of 2.68. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.22% to 0.33% per month and have t-statistics exceeding 2.29 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.10, with a t-statistic of 4.91 on the UMD factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 635 stocks and an average market capitalization of at least \$2,550 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 3.15. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -4-32bps/month. The lowest return, (-4 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.52. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BSV trad-

ing strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the BSV strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BSV, as well as average returns and alphas for long/short trading BSV strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the BSV strategy achieves an average return of 32 bps/month with a t-statistic of 2.60. Among these large cap stocks, the alphas for the BSV strategy relative to the five most common factor models range from 28 to 43 bps/month with t-statistics between 2.29 and 3.51.

5 How does BSV perform relative to the zoo?

Figure 2 puts the performance of BSV in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the BSV strategy falls in the distribution. The BSV strategy’s gross (net) Sharpe ratio of 0.44 (0.39) is greater than 84% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BSV strategy (red line).² Ignoring trading costs, a \$1 invested in the BSV strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$1.79 which ranks the BSV strategy in the top 4% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BSV strategy would have yielded \$1.46 which ranks the BSV strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016b\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BSV relative to those. Panel A shows that the BSV strategy gross alphas fall between the 51 and 77 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BSV strategy has a positive net generalized alpha for five out of the five factor models. In these cases BSV ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BSV add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of BSV with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BSV or at least to weaken the power BSV has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV}BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BSV and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BSV signal in these Fama-MacBeth regressions

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

exceed 1.75, with the minimum t-statistic occurring when controlling for Advertising Expense. Controlling for all six closely related anomalies, the t-statistic on BSV is 2.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the BSV strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BSV strategy earns alphas that range from 20-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.09, which is achieved when controlling for Advertising Expense. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BSV trading strategy achieves an alpha of 19bps/month with a t-statistic of 1.99.

7 Does BSV add relative to the whole zoo?

Finally, we can ask how much adding BSV to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the BSV signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$36.86, while \$1 investment in the combination strategy

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BSV is available.

that includes BSV grows to \$34.13.

8 Conclusion

This study provides compelling evidence that Balance Sheet Velocity (BSV) represents a statistically and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BSV-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.39, which remains robust after controlling for established risk factors. Most notably, BSV maintains its predictive power with a monthly alpha of 25 basis points (t -statistic = 2.64) relative to the Fama-French five-factor model plus momentum, indicating that it captures unique information not subsumed by existing factors.

The economic significance of our findings is further reinforced by the strategy’s ability to generate positive alpha even when controlling for six closely related accounting-based signals from the factor zoo. The persistence of a 19 basis point monthly alpha (t -statistic = 1.99) after adjusting for factors such as accruals, leverage, and changes in operating liabilities suggests that BSV contains distinct information about firm fundamentals that is valuable for return prediction. These results have important practical implications for both academic researchers and practitioners, as they identify a novel source of alpha that appears robust to implementation costs and survives stringent empirical tests.

However, several limitations warrant consideration. First, while our results are statistically significant, the marginal significance (t -statistic = 1.99) of the alpha relative to the extended factor model suggests that BSV’s incremental contribution may be modest when sophisticated investors already exploit related accounting anoma-

lies. Second, the study period and market conditions examined may not fully capture the signal's performance across different market regimes or in international markets. Third, the practical implementation of BSV strategies may face capacity constraints and market impact costs not fully captured in our analysis.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying BSV's predictive power could provide valuable insights into why this signal generates alpha. Understanding whether BSV captures information about operational efficiency, financial constraints, or managerial quality would enhance our theoretical understanding of the anomaly. Second, examining the signal's performance in international markets would test its robustness across different accounting standards and market structures. Third, exploring potential enhancements through machine learning techniques or combinations with other signals could improve the strategy's risk-return profile. Finally, investigating the time-varying nature of BSV's predictive power and its relationship with macroeconomic conditions could provide insights into when the signal is most effective. Despite these areas for further investigation, our findings establish Balance Sheet Velocity as a valuable addition to the constellation of factors explaining cross-sectional return variation.

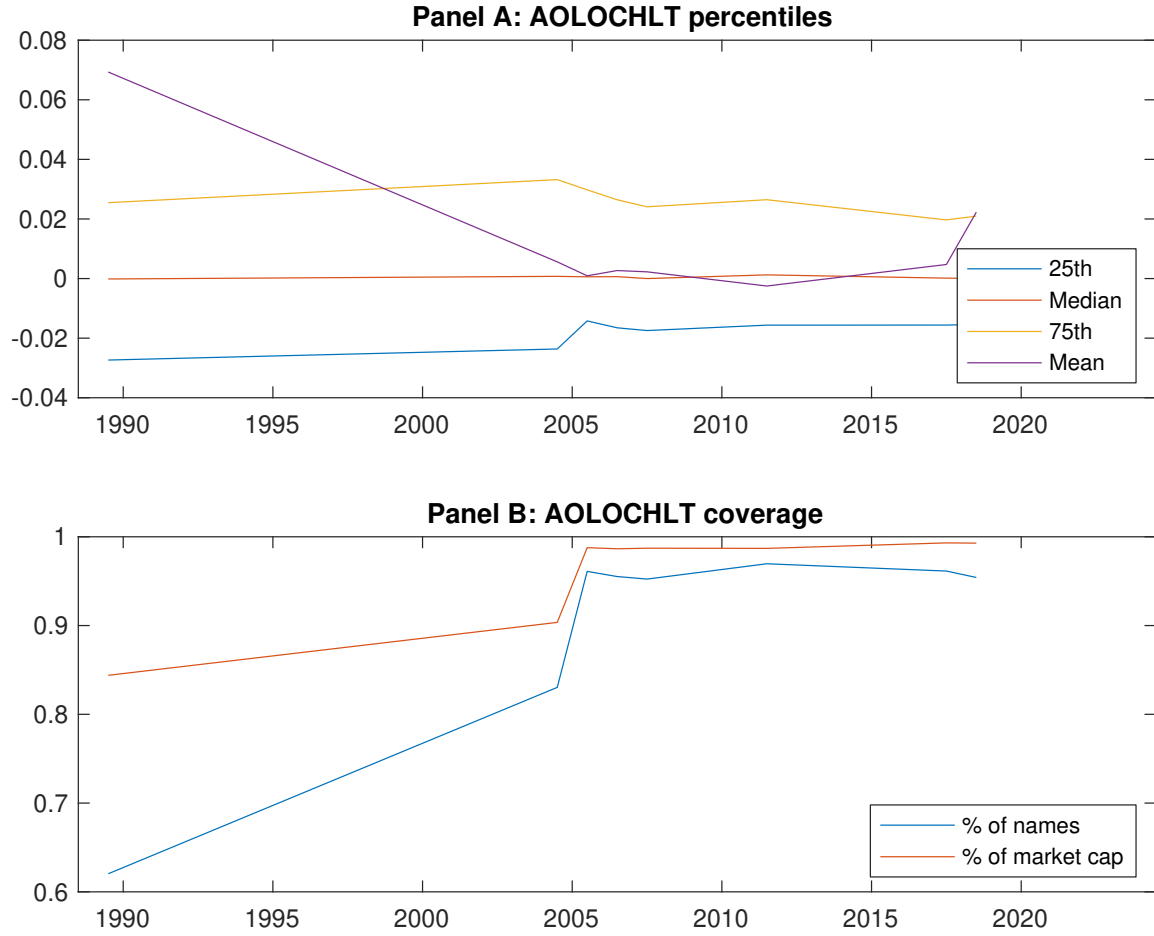


Figure 1: Times series of BSV percentiles and coverage.
This figure plots descriptive statistics for BSV. Panel A shows cross-sectional percentiles of BSV over the sample. Panel B plots the monthly coverage of BSV relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BSV. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on BSV-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.60 [2.58]	0.76 [3.61]	0.68 [3.47]	0.73 [3.50]	0.86 [3.68]	0.26 [2.68]
α_{CAPM}	-0.19 [-2.91]	0.05 [0.76]	0.03 [0.38]	0.02 [0.32]	0.07 [1.01]	0.26 [2.68]
α_{FF3}	-0.17 [-2.67]	0.01 [0.26]	-0.02 [-0.37]	-0.01 [-0.17]	0.12 [2.06]	0.29 [3.05]
α_{FF4}	-0.11 [-1.78]	0.03 [0.49]	-0.01 [-0.10]	-0.01 [-0.18]	0.11 [1.78]	0.22 [2.29]
α_{FF5}	-0.11 [-1.74]	-0.05 [-0.85]	-0.12 [-2.11]	-0.06 [-1.23]	0.22 [3.80]	0.33 [3.41]
α_{FF6}	-0.06 [-1.03]	-0.03 [-0.56]	-0.10 [-1.78]	-0.06 [-1.16]	0.20 [3.44]	0.26 [2.77]
Panel B: Fama and French (2018) 6-factor model loadings for BSV-sorted portfolios						
β_{MKT}	1.01 [64.51]	1.01 [68.91]	0.94 [66.78]	1.00 [75.86]	1.01 [69.41]	-0.00 [-0.06]
β_{SMB}	0.08 [3.43]	-0.02 [-0.88]	-0.01 [-0.39]	-0.06 [-3.36]	-0.01 [-0.40]	-0.08 [-2.49]
β_{HML}	-0.09 [-3.32]	0.06 [2.50]	0.12 [4.92]	0.06 [2.43]	-0.12 [-4.79]	-0.03 [-0.74]
β_{RMW}	-0.11 [-3.88]	0.08 [3.04]	0.18 [6.92]	0.09 [3.81]	-0.14 [-5.11]	-0.02 [-0.57]
β_{CMA}	0.01 [0.22]	0.14 [3.83]	0.11 [2.95]	0.07 [2.07]	-0.20 [-5.34]	-0.21 [-3.39]
β_{UMD}	-0.07 [-5.25]	-0.03 [-2.01]	-0.03 [-2.22]	-0.00 [-0.39]	0.03 [2.42]	0.10 [4.91]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1070	644	682	635	1031	
me (\$10 ⁶)	2550	2979	3241	3598	3980	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the BSV strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [2.68]	0.26 [2.68]	0.29 [3.05]	0.22 [2.29]	0.33 [3.41]	0.26 [2.77]
Quintile	NYSE	EW	0.19 [3.15]	0.20 [3.39]	0.20 [3.44]	0.14 [2.45]	0.20 [3.30]	0.15 [2.50]
Quintile	Name	VW	0.29 [2.41]	0.30 [2.47]	0.32 [2.79]	0.25 [2.14]	0.36 [3.02]	0.29 [2.46]
Quintile	Cap	VW	0.30 [2.88]	0.28 [2.65]	0.33 [3.31]	0.26 [2.58]	0.40 [3.99]	0.33 [3.35]
Decile	NYSE	VW	0.35 [2.45]	0.35 [2.39]	0.40 [2.85]	0.29 [2.12]	0.47 [3.29]	0.37 [2.66]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.38]	0.23 [2.34]	0.25 [2.66]	0.21 [2.22]	0.29 [3.02]	0.25 [2.64]
Quintile	NYSE	EW	-0.04 [-0.52]					
Quintile	Name	VW	0.25 [2.13]	0.25 [2.10]	0.28 [2.36]	0.23 [1.97]	0.31 [2.60]	0.27 [2.25]
Quintile	Cap	VW	0.28 [2.63]	0.27 [2.47]	0.31 [3.07]	0.26 [2.65]	0.37 [3.74]	0.33 [3.36]
Decile	NYSE	VW	0.32 [2.19]	0.30 [2.00]	0.33 [2.38]	0.27 [1.96]	0.40 [2.83]	0.34 [2.45]

Table 3: Conditional sort on size and BSV

This table presents results for conditional double sorts on size and BSV. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BSV. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BSV and short stocks with low BSV. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	BSV Quintiles					BSV Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.50 [1.43]	0.69 [2.13]	0.73 [2.48]	0.80 [2.64]	0.68 [1.93]	0.18 [1.74]	0.17 [1.62]	0.16 [1.52]	0.11 [1.01]	0.18 [1.68]	0.14 [1.26]
	(2)	0.67 [2.02]	0.68 [2.40]	0.82 [3.01]	0.82 [2.90]	0.67 [2.14]	0.00 [0.00]	0.07 [0.64]	0.07 [0.67]	0.06 [0.58]	0.05 [0.45]	0.04 [0.37]
	(3)	0.76 [2.55]	0.64 [2.47]	0.75 [2.93]	0.81 [3.15]	0.81 [2.76]	0.05 [0.52]	0.05 [0.48]	0.06 [0.61]	0.06 [0.57]	0.15 [1.40]	0.14 [1.27]
	(4)	0.73 [2.77]	0.79 [3.29]	0.72 [3.07]	0.84 [3.62]	0.89 [3.20]	0.16 [1.53]	0.14 [1.28]	0.20 [2.03]	0.13 [1.33]	0.33 [3.42]	0.27 [2.77]
	(5)	0.57 [2.53]	0.77 [3.76]	0.70 [3.56]	0.75 [3.58]	0.89 [3.73]	0.32 [2.60]	0.30 [2.34]	0.35 [2.86]	0.28 [2.29]	0.43 [3.51]	0.37 [2.99]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	BSV Quintiles					BSV Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	451	451	450	449	449	47	55	55	54	51	
	(2)	133	133	132	133	133	91	93	94	95	93	
	(3)	90	90	90	90	90	158	159	159	162	160	
	(4)	74	74	74	74	74	326	342	341	343	341	
(5)	66	66	66	66	66	2612	2266	2430	2939	2882		

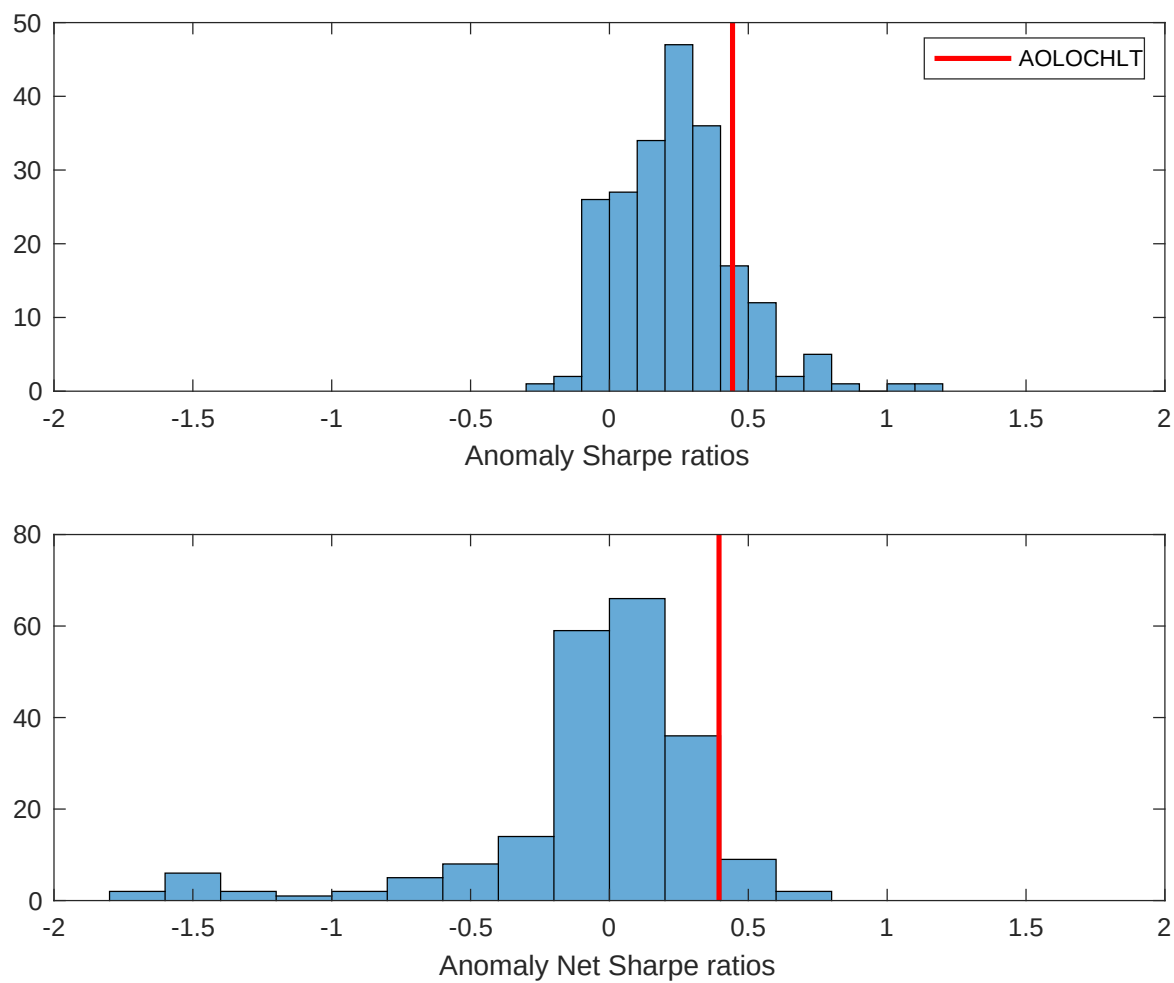


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BSV with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

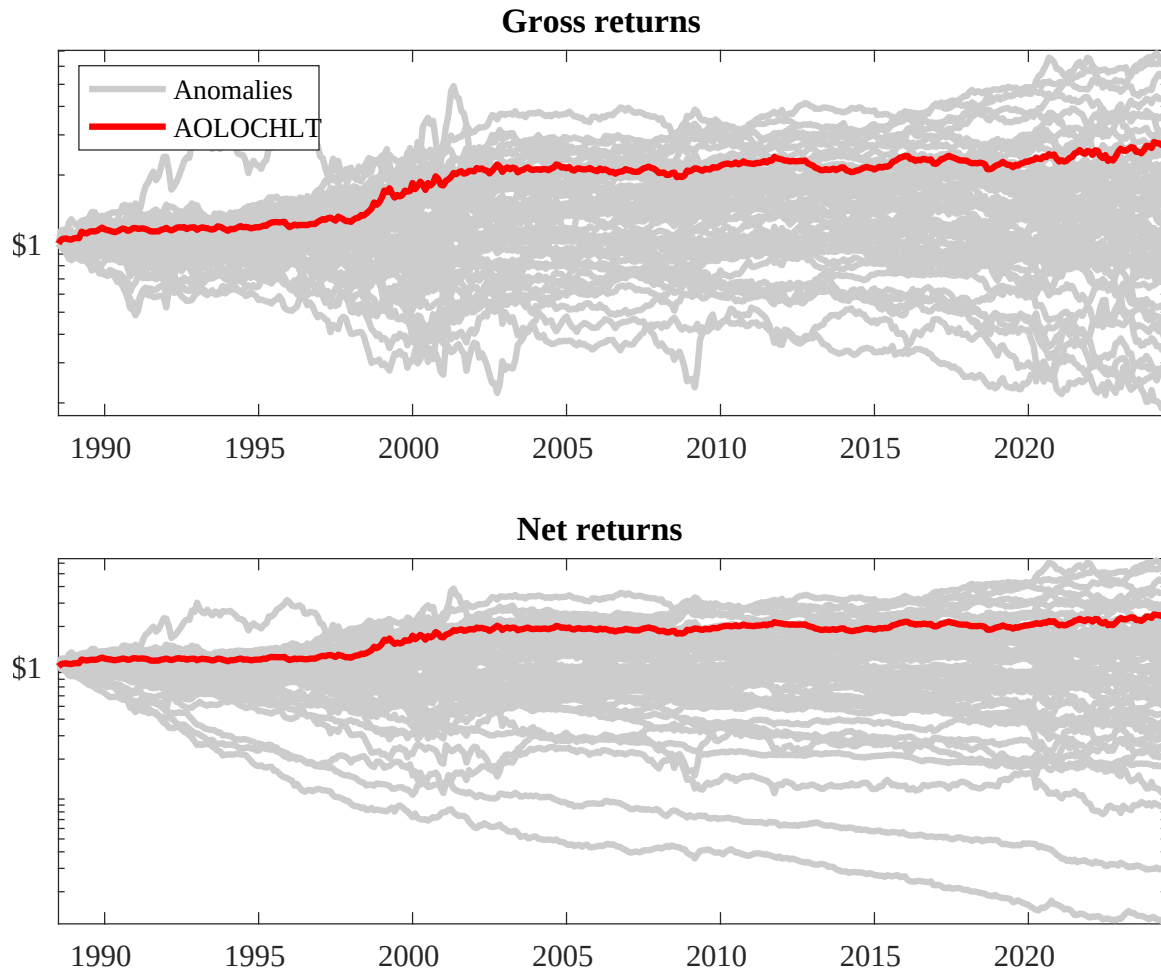
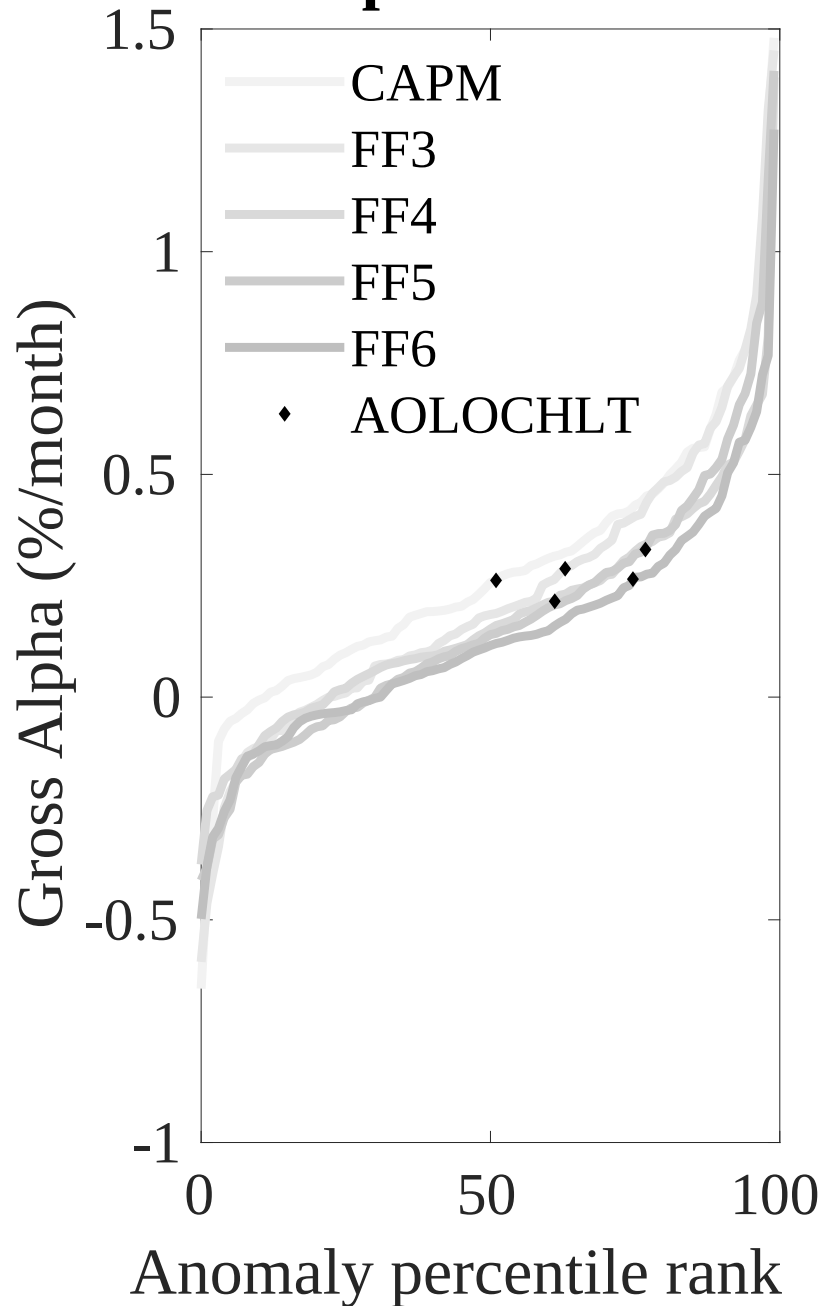


Figure 3: Dollar invested.

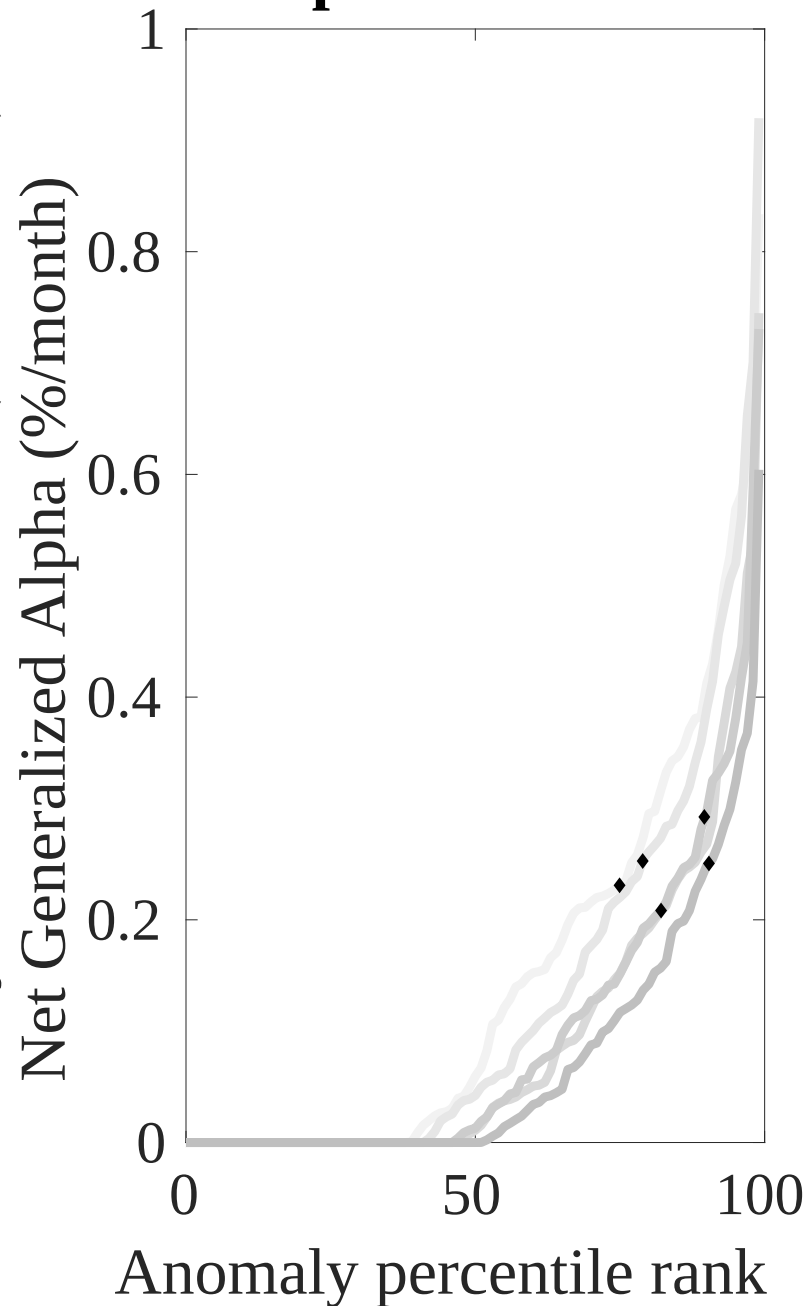
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BSV trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



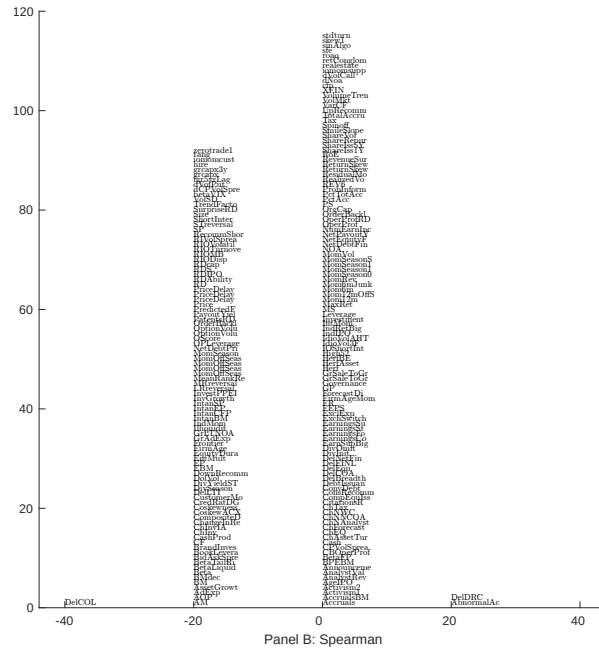
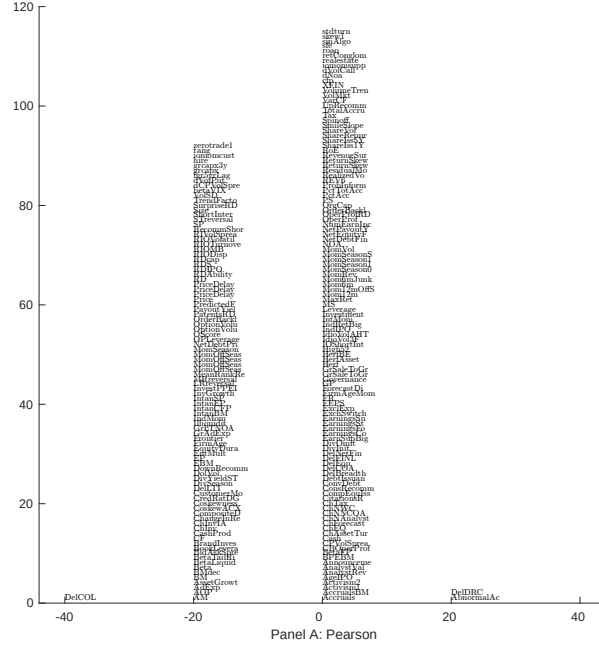


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with BSV. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

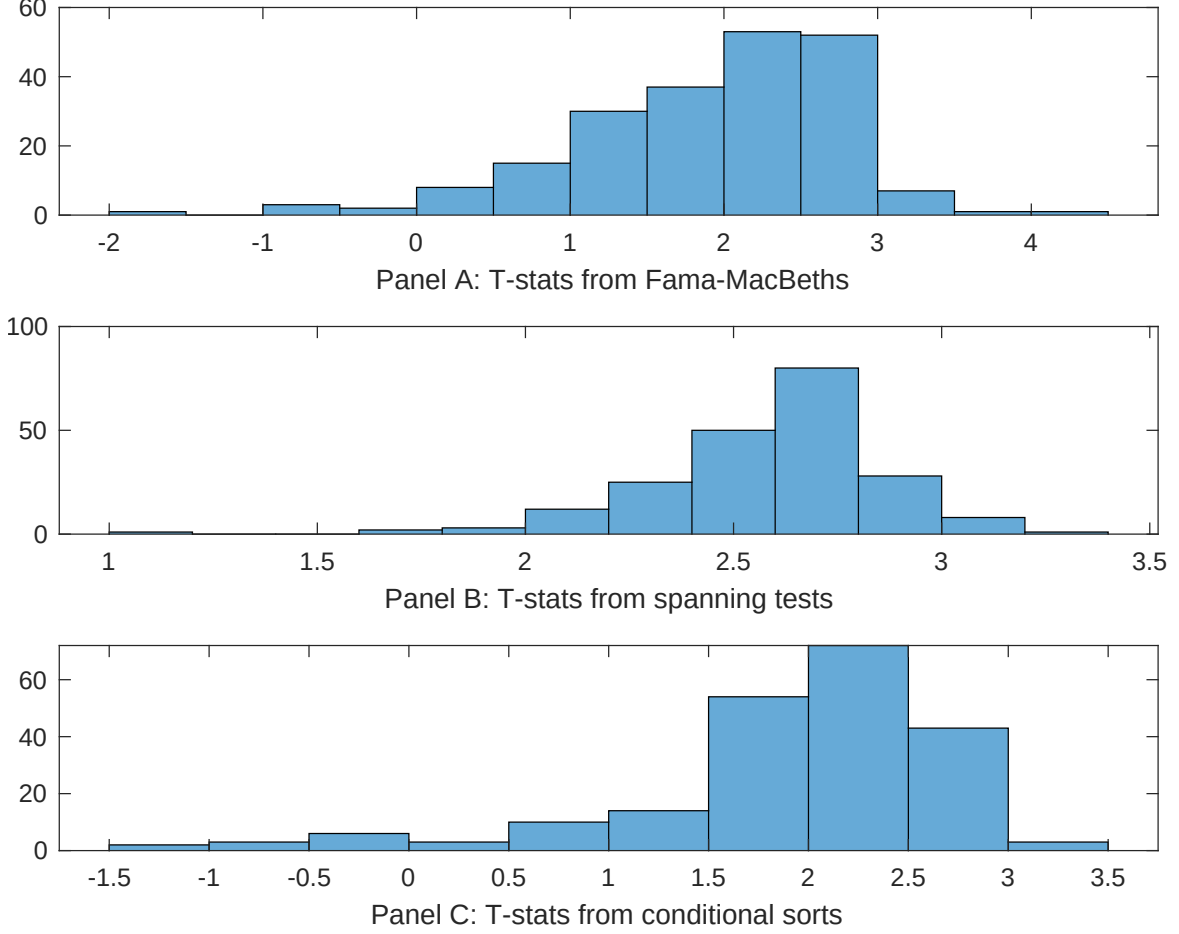


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of BSV conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BSV} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BSV}BSV_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BSV,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BSV. Stocks are finally grouped into five BSV portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BSV trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on BSV. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BSV}BSV_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.11 [3.99]	0.11 [3.58]	0.11 [3.88]	0.94 [3.32]	0.11 [3.88]	0.11 [3.91]	0.12 [4.22]
BSV	0.64 [3.11]	0.51 [2.57]	0.40 [1.89]	0.52 [1.75]	0.45 [2.10]	0.39 [2.05]	0.71 [2.43]
Anomaly 1	0.21 [4.69]						0.24 [3.38]
Anomaly 2		0.14 [1.45]					0.19 [1.72]
Anomaly 3			0.51 [1.90]				0.76 [2.46]
Anomaly 4				0.56 [0.08]			-0.14 [-0.21]
Anomaly 5					0.56 [0.19]		-0.49 [-1.35]
Anomaly 6						0.14 [5.62]	0.14 [4.73]
# months	432	432	432	427	432	432	427
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BSV trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BSV} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in current operating liabilities, Book leverage (annual), Total accruals, Advertising Expense, Abnormal Accruals, Change in Taxes. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.20 [2.09]	0.24 [2.49]	0.28 [2.91]	0.28 [2.83]	0.22 [2.24]	0.27 [2.83]	0.19 [1.99]
Anomaly 1	-22.95 [-5.21]						-12.80 [-2.50]
Anomaly 2		15.56 [4.46]					6.82 [1.74]
Anomaly 3			-15.31 [-3.46]				-5.20 [-1.07]
Anomaly 4				-2.61 [-0.88]			0.58 [0.20]
Anomaly 5					15.74 [3.52]		10.15 [2.26]
Anomaly 6						17.60 [4.57]	12.09 [3.06]
mkt	0.29 [0.12]	1.02 [0.43]	0.09 [0.04]	-0.26 [-0.11]	-1.90 [-0.79]	-1.93 [-0.81]	-1.90 [-0.79]
smb	-10.32 [-3.05]	-11.54 [-3.33]	-9.35 [-2.73]	-7.09 [-1.91]	-9.22 [-2.69]	-8.26 [-2.45]	-12.30 [-3.39]
hml	8.20 [1.77]	8.89 [1.81]	-4.51 [-1.08]	-1.94 [-0.44]	-0.98 [-0.23]	-0.48 [-0.12]	10.84 [2.04]
rmw	-1.87 [-0.44]	1.26 [0.29]	-5.69 [-1.29]	-1.85 [-0.41]	-0.38 [-0.09]	-2.59 [-0.61]	-0.74 [-0.17]
cma	-9.12 [-1.43]	-20.72 [-3.45]	-10.57 [-1.56]	-20.14 [-3.19]	-17.61 [-2.87]	-19.06 [-3.17]	-7.15 [-1.01]
umd	9.24 [4.42]	9.87 [4.70]	9.06 [4.22]	9.29 [3.82]	9.67 [4.55]	5.83 [2.52]	5.65 [2.30]
# months	432	432	432	428	432	432	428
$\bar{R}^2(\%)$	18	17	15	13	15	17	22

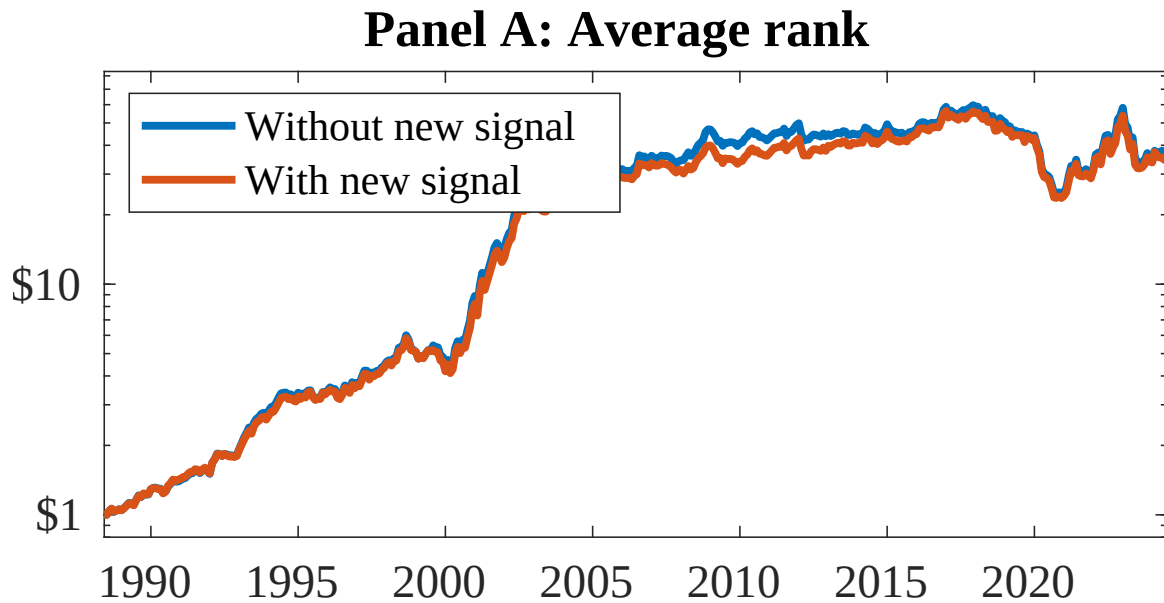


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as BSV. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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